

These are just some of the leaks that happened in January this year. Companies like Kroger and T-Mobile were targeted by hackers in February. Microsoft Exchange and SITA (the latter supports 90 per cent of the world's airlines) were targeted in March.

Major companies in the industry with their endless supply of financing, manpower and resources are unable to save their data from being hacked. Looking at the statistics above, it is evident that if you are running a Web application that stores customer data in any form, then you need network security.

I have aggregated a list of what, in my opinion, are the top 10 open source network security tools that you could make use of. The list is not arranged in any order of importance or potential. Each tool brings its own set of features. Feel free to pick and choose whatever tools work best for you.

10. Zed Attack Proxy (ZAP)

A project developed and maintained by OWASP, ZAP is one of the most popular open source network security tools in the industry. It is multi-platform based with a user-friendly GUI and a CLI for advanced developers. With its latest version (v2.10.0) released in December 2020, ZAP has an active developer community with good documentation and support. The major highlights of ZAP are:

- Active and vibrant developer community on GitHub
- SQL injection vulnerability test
- XSS injection vulnerability test
- Easy to use and intuitive for beginners
- Automated scanning of files
- Uses a REST based API
- Written in Java programming language

9. Wfuzz

Wfuzz is a brute forcing tool developed for Web applications. It brute forces the GET and POST parameters for checking different types of injection vulnerabilities. Its latest version (v3.1.0) was released in November 2020 and it has an active developer community. Unfortunately, it doesn't have a GUI and can only be accessed using the terminal. The major highlights of Wfuzz are:

- Multiple injection point capabilities with multiple dictionaries
- LDAP injection vulnerability test
- SQL injection vulnerability test
- XSS injection vulnerability test
- HEAD scan
- Multi-threading support
- Written in Python programming language

8. Nikto2

Nikto is an open source Web server scanner. It helps you scan your Web server, testing for more than 6,700 potentially harmful or dangerous files or programs. It also checks for outdated versions of roughly 1200+ servers and version specific problems on more than 250 servers. It does not have a GUI and can only be accessed using the terminal. Its latest version (v2.10.0) was released in December 2020 and it has an active developer community. The major highlights of Nikto2 are:

- Full HTTP proxy support
- SSL support
- Save reports in HTML, XML and or plain text
- Easily identify installed software in the servers
- Maximum execution time per target
- Reporting of unusual headers
- Well defined and thorough documentation

7. OpenVAS

Developed and maintained by the Greenbone network since 2009, OpenVAS stands for Open Vulnerability Assessment Scanner. It is an open source tool used for testing vulnerabilities in a Web application. With its latest version (v20.8.1) out in February 2021, OpenVAS is an actively maintained open source tool with good community support. Major highlights are:

- Unauthenticated / authenticated testing
- High / low level Internet and industrial protocols
- Performance tuning for large scale scan
- Powerful internal programming language
- Accompanied by a vulnerability tests feed containing over 80,000 tests
- Compatible with Greenbone vulnerability management suite
- XML based stateless request-response

6. Grabber

Grabber is a fun little penetration testing tool with the ability to test many different vulnerabilities. Written in Python, this open source tool requires you to have some background knowledge in understanding vulnerabilities, as it only shows you what vulnerabilities your Web app has but not how to resolve them. There is no standardised version mentioned for this tool on its GitHub repository and community support might be a problem. The major highlights of Grabber are:

- Cross-site scripting vulnerability test
- SQL injection vulnerability test
- Simple AJAX check
- Crystal ball testing for PHP applications